

Data Analysis and Visualization

PART III: Analyzing data, Interpreting the analyses, Communicating the results statistics

Module: Inferences about population proportion and central values

Point estimation and confidence interval

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Statistics Fundamentals

For your reference: <https://www.youtube.com/statquest>

- Video: Population and Estimated Parameters, Clearly Explained!!!
- Video: Bootstrapping Main Ideas!!!
- Video: The Central Limit Theorem, Clearly Explained!!!
- Video: Standard Deviation vs Standard Error, Clearly Explained!!!
- Video: Confidence Intervals, Clearly Explained!!!

Introduction

- The objective of statistics is to make inferences about a population based on information contained in a sample.
- Populations are characterized by numerical descriptive measures called *parameters*.
- Typical population parameters are the mean μ , the standard deviation σ , and a proportion p , etc.
- Methods for making inferences about parameters fall into one of two categories:
 - ▶ Estimate the value of the population parameters of interest
 - ▶ Test a hypothesis about the value of the parameters

Section:

Estimating of the Population Proportion

- Point estimation
- Confidence interval
- Hypothesis testing (week 7)

Point estimation of the population proportion p

- The first step in statistical inference is point estimation, in which we compute a single value (statistic) from the sample data to estimate a population parameter.
- The sample proportion, denoted by $\hat{p}$, is a point estimate of the population proportion p :

$$\hat{p} = \frac{\text{the number of successes in the sample}}{\text{the sample size}}$$

It is the "best estimate" (unbiased point estimate) for the population proportion in the sense that expectation of estimator is approximated to population parameter.

Estimating the population proportion with confidence interval

Example: A study on the proportion of parents who use a car seat for their toddler when traveling.

Analysis:

- Population: parents with toddlers.
- Parameter of interest: population proportion.
- Objective: Construct a 95% confidence interval for the proportion of parents who use a car seat when traveling with a toddler.

Observed information (data):

- Sample data: 659 parents with toddlers.
- Survey result: 540 parents reported using a car seat.

Calculating the 95% Confidence Interval

- Point estimation for population proportion is sample proportion

$$\hat{p} = \frac{540}{659} = 0.85$$

- Want to construct a 95% confidence interval to estimate the population proportion.
 - ▶ Define $X_1, X_2, \dots, X_n$: 1 if parent uses car seat with probability $\hat{p}$, 0 otherwise with probability $1 - \hat{p}$
 - ▶ Then, $X_1, X_2, \dots, X_n$ is independent identically distributed with $X_i \sim \text{Bernoulli}(\hat{p})$.
 - ▶ $E[X_i] = \hat{p}$, $\text{Var}(X_i) = \hat{p}(1 - \hat{p})$ for all $i = 1, \dots, n$
 - ▶ $n = 659$ is sample size or the number of data point.

Calculating the 95% Confidence Interval

- Each observation follows a Bernoulli distribution, the sampling distribution of the sample proportion $\hat{p}$ is approximately normal (Central Limit Theorem):

$$\hat{p} = \frac{X_1 + X_2 + \cdots + X_n}{n} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

where $\mu = E[X_i] = \hat{p} = 0.85$, and $\sigma^2 = \text{Var}(X_i) = \hat{p}(1 - \hat{p})$

- Using the Central Limit Theorem, the 95% confidence interval is:

$$\begin{aligned}\hat{p} \pm Z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} &= 0.85 \pm 1.96 \sqrt{\frac{0.85 \cdot (1 - 0.85)}{659}} \\ &= (0.8227, 0.8773)\end{aligned}$$

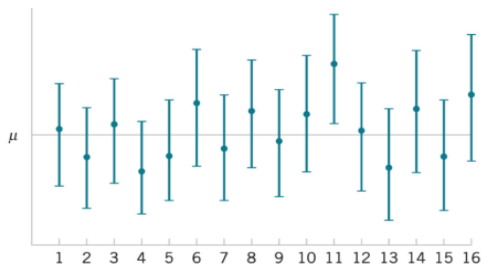
Conclusion and Interpretation

- **Conclusion:**

- ▶ The estimated proportion of parents who use car seats when traveling with toddlers is 85%.
- ▶ We are 95% confident that the true population proportion lies between 82.27% and 87.73%.

- **Interpretation:**

- ▶ If we repeat this sampling process many times, approximately 95% of the resulting confidence intervals will contain the true population proportion.



Conclusion: Estimating Population Proportion with Confidence Intervals

- A random sample is a subset of the population selected such that each member of the population has an equal chance of being chosen.
- A random sample of size n should be large enough such as both np and $n(1 - p)$ are greater than 10.
- Confidence Interval for Population Proportion:

$$\hat{p} \pm Z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- ▶ $\hat{p}$ is the sample proportion
- ▶ n is the sample size
- ▶ Z^* is the critical value corresponding to the confidence level
- ▶ Confidence Levels and Corresponding Z^* Values: 90% & ± 1.645 ; 95% & ± 1.96 ; 98% & ± 2.326 ; 99% & ± 2.576

Notes on Interpreting Confidence Intervals

- When a confidence interval is constructed, it does not mean the population parameter definitely lies in that range — rather, it means we are confident based on the method.
- A 95% confidence level means that if we took many random samples and built a confidence interval for each, about 95% of those intervals would contain the true population proportion.
- Once a specific confidence interval is constructed from a sample, it either contains the true proportion or it doesn't — we don't know for sure in a single case.
- Higher confidence levels (like 99%) result in wider intervals. Lower confidence levels (like 90%) give narrower intervals.
- Confidence intervals are a core tool in inferential statistics to estimate population parameters using sample data.

Section:

Estimating Population Proportion Difference with Two Independent Samples

- Point estimation
- Confidence interval
- Hypothesis testing (week 7)

Estimating Population Proportion Difference with Two Independent Samples

- **Example:** Consider a study comparing the proportion of children (aged 6–18) who attend swimming class in urban vs. rural areas.
- Define:
 - ▶ Population: All children aged 6–18 in the country.
 - ▶ Group 1: Urban children aged 6–18 (Population 1)
 - ▶ Group 2: Rural children aged 6–18 (Population 2)
- Objective: Estimate the difference in swimming class attendance proportions between urban and rural children.
- Let population proportions difference be:

$$p_1 - p_2$$

Sample Statistics and Problem Statement

- Sample results:
 - ▶ 543 out of 988 urban children (aged 6–18) attend swimming class.
 - ▶ 91 out of 247 rural children (aged 6–18) attend swimming class.
- How accurately can we estimate the difference in school attendance proportions between urban and rural children?
- How to construct a 95% confidence interval for the difference $p_1 - p_2$.
- Denote:
 - ▶ p_1 is the first population proportion (urban)
 - ▶ n_1 is the sample size of the first population (urban)
 - ▶ p_2 is the second population proportion (rural)
 - ▶ n_2 is the sample size of the second population (rural)

Constructing the 95% Confidence Interval

- Step 1: Define sample proportions

$$\hat{p}_1 = \frac{543}{988} \approx 0.55, \quad \hat{p}_2 = \frac{91}{247} \approx 0.37, \quad \hat{p}_1 - \hat{p}_2 \approx 0.18$$

- Step 2: Standard error of difference (WHY?)

$$SE = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

- Step 3: 95% Confidence Interval is

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \cdot SE$$

where $z^* = 1.96$ for 95% confidence. Hence, the confidence interval is

$$0.18 \pm 0.0677 \Rightarrow (0.1123, 0.2477)$$

Interpretation and Assumptions

- Interpretation:

- ▶ With 95% confidence, the proportion of urban children attending swimming class is estimated to be between 11.23% and 24.77% higher than that of rural children.

- Notes:

- ▶ The two samples must be independent random samples.
- ▶ Sample sizes must be large enough (normal approximation condition):

$$n_1\hat{p}_1, \quad n_1(1 - \hat{p}_1), \quad n_2\hat{p}_2, \quad n_2(1 - \hat{p}_2) > 10$$

This ensures reliable results using the normal distribution.

Section: Estimating of the Population Mean

- Point estimation
- Confidence interval
- Hypothesis testing (week 7)

Estimating the Population Mean with a Confidence Interval

Example: Estimate the average distance that people can perform a standing long jump (i.e. cartwheel, in inches).

Analysis:

- Population: All individuals adults
- Parameter of interest: Population mean of cartwheel distance
- Goal: Estimate the population mean distance μ (true mean) with 95% confidence interval

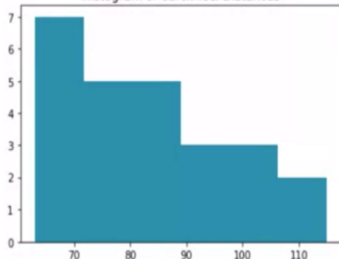
Observed data:

- Suppose we randomly select 25 people and measure their long jump distances.
- The sample mean is 82.48 inches, and sample standard deviation is 15.06 inches. (figure next slide)

Dataset and Initial Analysis

	ID	Age	Gender	GenderGroup	Glasses	GlassesGroup	Height	Wingspan	CWDistance	Complete	CompleteGroup	Score
0	1	56	F	1	Y	1	62.0	61.0	79	Y	1	7
1	2	26	F	1	Y	1	62.0	60.0	70	Y	1	8
2	3	33	F	1	Y	1	66.0	64.0	85	Y	1	7
3	4	39	F	1	N	0	64.0	63.0	87	Y	1	10
4	5	27	M	2	N	0	73.0	75.0	72	N	0	4

Histogram of Cartwheel Distances



Cartwheel Distance Traveled (in inches)

```
df.describe()["CWDistance"]
```

```
count    25.000000
mean      82.480000
std       15.058552
min       63.000000
25%       70.000000
50%       81.000000
75%       92.000000
max       115.000000
Name: CWDistance, dtype: float64
```

Point estimation for population mean and std

- Point estimation: "best estimator" for population mean μ is sample mean

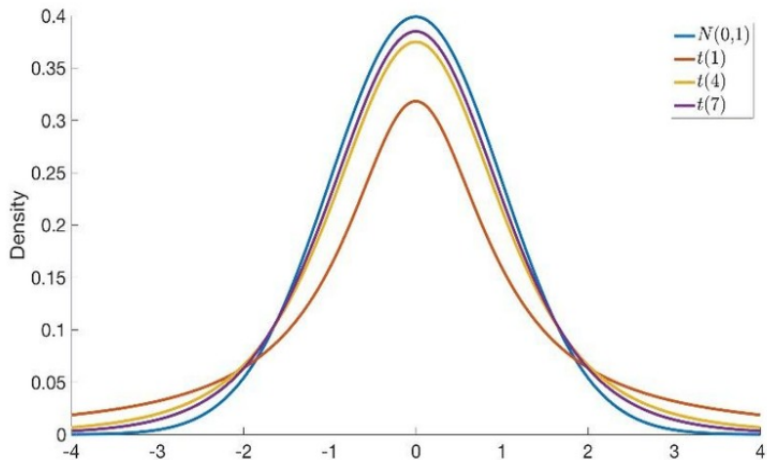
$$\hat{\mu} = \bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = 82.48$$

- Point estimation: estimator for population standard deviation μ is sample standard deviation

$$\hat{\sigma} = s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2} = 15.06$$

- Sample size $n = 25$, not large enough to apply Central Limit Theorem $\implies$ approximated population data by t -distribution

t -distribution



Confidence Interval formula:

$$\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

- $\bar{x}$: sample mean
- s : sample standard deviation
- n : sample size
- t^* : critical value from the t-distribution (degree of freedom $df = n - 1$)

Apply to our research, we have

- Confidence level is 95%, sample size is 25 $\implies t^* = 2.064$
- 95% Confidence interval of the population mean is

$$82.48 \pm 2.064 \cdot \frac{15.06}{\sqrt{25}}$$

(76.26, 88.70) (inches)

Conclusion and Interpretation of 95% Confidence

- **Conclusion (95% Confidence Interval)**

- ▶ With 95% confidence, the population mean cartwheel distance (μ) for all adults is estimated to be between 76.26 inches and 88.70 inches.
- ▶ (This interval represents a plausible range of values for the population parameter μ).

- **Interpretation of 95% Confidence**

- ▶ The 95% confidence level means that if we were to repeat this sampling and interval estimation process many times, we would create a large number of 95% confidence intervals.
- ▶ We would expect that approximately 95% of all these intervals created would contain the true population mean cartwheel distance (μ).

Alternative approach: Deriving the CI

- $X_1, X_2, \dots, X_n$ i.i.d with $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$
- CLT say, sampling distribution of the sample mean ($\bar{X}$):

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ (for large } n, \text{ i.e., } n \geq 30)$$

- Standardized Test Statistic (Z):

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

- - ▶ We find the critical value z^* (e.g., $z^* = 1.96$ for 95% CI, since $1 - \alpha = 95\%$).

$$P\left(-z^* \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z^*\right) = 1 - \alpha$$

- - ▶ **Final Confidence Interval Formula for μ :**

$$P\left(\bar{X} - z^* \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z^* \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Section: Estimating the Difference in Population Means Between Two Independent Populations

- Point estimation
- Confidence interval
- Hypothesis testing (week 7)

Estimating the Difference in Population Means Between Two Independent Populations

- Research Problem Example:
 - ▶ Consider all adults aged 18–29 living in Vietnam. Is there a significant difference in the average Body Mass Index (BMI) between males and females in this age group?
- Analysis:
 - ▶ Population: All adults aged 18–29 currently living in Vietnam.
 - ▶ Population parameters of interest: $\mu_1 - \mu_2$
 - ▶ μ_1 : mean BMI of males
 - ▶ μ_2 : mean BMI of females

Data summary and visualization

Dữ liệu:

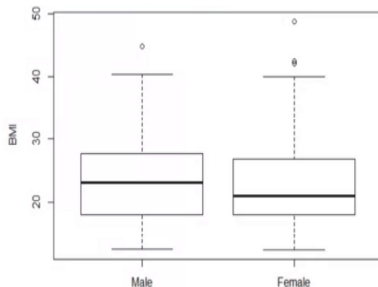
Gender	BMI	Race	Age 18-29
1	19.9	1	1
2	17.0	1	1
2	26.7	1	1
1	25.6	1	1
...	...	...	...

Gender (giới tính): 1 - nam
2 - nữ

Chú ý: dữ liệu thu thập
độc lập và đại diện được
cho tổng thể

Mô tả dữ liệu:

	Male	Female
Mean	23.57	22.83
St. Dev.	6.24	6.43
Min	12.5	12.4
Max	44.9	48.8
n	258	239



Estimating the difference in population means

- Given data from two samples:
 - Sample mean BMI of males follows approximately a normal distribution: $\bar{X}_1 \sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right)$
 - Sample mean BMI of females also follows approximately a normal distribution: $\bar{X}_2 \sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$
- Since the two samples are independent, the difference in sample means follows:

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

- Standard deviation of the difference:

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- When $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ are unknown $\rightarrow$ we estimate from the sample.

Estimating the difference in population means

- Estimate $\mu_1 - \mu_2$ using the difference between sample means: $\bar{X}_1 - \bar{X}_2$.
- Estimate σ_1^2, σ_2^2 using the sample variances s_1^2, s_2^2 .
- The confidence interval is defined as:

$$(\bar{X}_1 - \bar{X}_2) \pm t^* \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where t^* is determined by the degrees of freedom df using:

$$df = \min(n_1 - 1, n_2 - 1)$$

and then t^* obtained from the t -distribution table at 95% confidence.

Constructing CI for difference between two population means

When constructing confidence intervals for the difference between two population means ($\mu_1 - \mu_2$) with independent samples, there are two cases:

Case 1 (General) - Unpooled Confidence Interval:

$$(\bar{X}_1 - \bar{X}_2) \pm t^* \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- t^* is the critical value from the t -distribution
- Degrees of freedom (df): $\min(n_1 - 1, n_2 - 1)$
- For 95% CI and large samples: $t^* \approx z^* = 1.96$
- $\bar{X}_1, \bar{X}_2$: sample means
 s_1^2, s_2^2 : sample variances
 n_1, n_2 : sample sizes

Constructing CI for difference between two population means

Case 2 - Pooled Confidence Interval (special case): Use when we assume equal population variances i.e., $\sigma_1^2 = \sigma_2^2$

$$(\bar{X}_1 - \bar{X}_2) \pm t^* \times \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

where the pooled variance is:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

- Degrees of freedom: $df = n_1 + n_2 - 2$
- Use the same process to look up t^* from t -distribution table
- If $n_1, n_2 \geq 30$, then use $t^* \approx z^*$

CI Calculation for different populations

The task is to compute the confidence interval for the difference in means using the following formula:

$$(\bar{X}_1 - \bar{X}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Given data:

$$\text{Male: } \bar{X}_1 = 23.57, s_1 = 6.24, n_1 = 258$$

$$\text{Female: } \bar{X}_2 = 22.83, s_2 = 6.43, n_2 = 239$$

Substitute the values into the formula:

$$(23.57 - 22.83) \pm 1.96 \times \sqrt{\frac{6.24^2}{258} + \frac{6.43^2}{239}}$$

The 95% confidence interval is:

$$(-0.3755, 1.8555) \text{ kg/m}^2$$

Pooled CI formula (equal variances assumption)

If there is enough evidence to assume that the variances of the two populations are equal, we use the **pooled confidence interval formula**:

$$(\bar{X}_1 - \bar{X}_2) \pm t^* \cdot \sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \right) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Given:

$$\text{Male: } \bar{X}_1 = 23.57, \quad s_1 = 6.24, \quad n_1 = 258$$

$$\text{Female: } \bar{X}_2 = 22.83, \quad s_2 = 6.43, \quad n_2 = 239$$

Substitute the values into the formula with $t^* = 1.96$, the 95% confidence interval is:

$$\Rightarrow (-0.3739, 1.8539) \text{ kg/m}^2$$

Interpretation of the Confidence Interval

Conclusion: With 95% confidence, the difference in mean BMI between all males and all females aged 18–29 in Vietnam is estimated to lie within the interval:

$$(-0.3755, 1.8555) \text{ kg/m}^2$$

What does 95% confidence mean?

If we were to repeat this sampling process many times, constructing a confidence interval each time, we expect that about 95% of those intervals would contain the true difference in population means

$$\mu_1 - \mu_2.$$

Lab 6

- Apply basic probability and statistics such as Bayes' formula, conditional probability, descriptive statistics, point estimation, parameter estimation of population parameters with confidence interval, etc to your research of interest.